

A Q4 plane-stress cantilever with the clamp imposed by Lagrange multipliers

Theory, implementation and analytical validation

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Outline

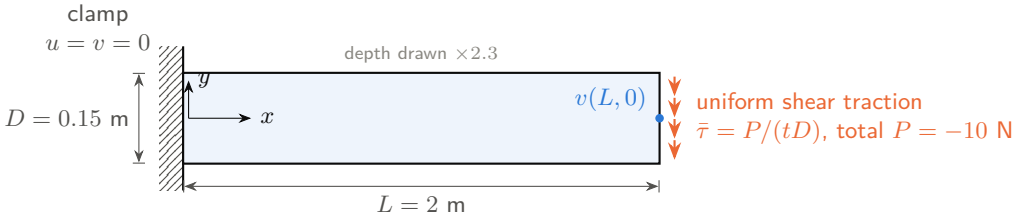
1. The problem and the validation plan
2. Theory
   - plane-stress elasticity, minimum potential energy
   - the Q4 element and Gauss quadrature
   - assembly and consistent loads
   - the clamp by Lagrange multipliers, scaling
   - beam theory, shear locking, convergence
3. Validation V1–V6
4. Conclusions and limitations

The rule followed throughout

Every number the code produces is paired with an **analytical counterpart**, computed from its formula, and every pairing is an automated test (109 tests). Every closed form is also derived symbolically (sympy).

Code: fem/ (element, assembly, Lagrange solve), checks/ (closed forms, beam theory, symbolic proofs), studies/ (figures), tests/.

The problem



- Plane stress, thickness $t = 0.1$ m, $E = 200$ GPa, $\nu = 0.3$
- Clamp: $u = v = 0$ at *every* node of $x = 0$, imposed by **Lagrange multipliers**
- Load: $P = -10$ N as a uniform shear traction on $x = L$
- Q4 elements, 2×2 Gauss, structured $n_x \times n_y$ meshes
- Nested meshes $3 \times 1 \to 6 \times 2 \to \cdots \to 384 \times 128$ (every element has $a/b = 4.44$)
- Outputs: tip deflection $v(L, 0)$, stresses, reactions $\mathbf{R} = -C^T \boldsymbol{\lambda}$

The validation plan: every result has an analytical counterpart

	The code computes	Analytical counterpart	Measured agreement
V1	1D bar, both ends prescribed (Lagrange)	$u = \frac{qx(L-x)}{2EA} + \frac{Ux}{L}$; support forces from $N = EAu'$	4.8×10^{-15}; $-\lambda = -700$, -300 N
V2	Q4 stiffness, 2×2 Gauss	exact integration for a rectangle $2a \times 2b$	2.3×10^{-16}; 3 zero eigenvalues
V3	uniform tension, distorted mesh	$u = \sigma_0 x/E$, $v = -\nu\sigma_0 y/E$ (patch test)	1.1×10^{-13}
V4	Q4 in pure bending	Cook Eq. 3.6-11	8.6×10^{-13}
V5	1×1 cantilever	complete hand solution (2 unknowns)	1.5×10^{-13}
V6	$3 \times 1 \ \ldots \ 384 \times 128$	statics, Clapeyron, minimum potential energy, beam theory, $O(h^2)$	round-off; -0.13% from Timoshenko

Agreement = largest relative difference between code and formula. Sources: figures/*.txt, regenerated by `python run_all.py`.

1. Plane-stress elasticity (strong form)

Equilibrium (no body force)

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0, \qquad \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} = 0$$

Strain–displacement (small strain)

$$\varepsilon_x = \frac{\partial u}{\partial x}, \quad \varepsilon_y = \frac{\partial v}{\partial y}, \quad \gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$$

Boundary conditions

$u = v = 0$ on $x = 0$; traction $(0, \bar{\tau})$ on $x = L$;
traction-free on $y = \pm D/2$

3D Hooke's law with $\sigma_z = \tau_{yz} = \tau_{zx} = 0$, solved for the stresses:

$$\varepsilon_x = \frac{\sigma_x - \nu \sigma_y}{E}, \qquad \varepsilon_y = \frac{\sigma_y - \nu \sigma_x}{E},$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G}, \qquad G = \frac{E}{2(1+\nu)}$$

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \underbrace{\frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}}_{D} \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix}$$

Why plane stress: the faces $z = \pm t/2$ are free, and beam theory (the reference) also assumes uniaxial stress, $\sigma_y = \sigma_z = 0$.

Cook Eq. 3.1-3. Code: fem/material.py. Proof: checks/derivations.py inverts the 3D compliance symbolically.

2. Principle of minimum total potential energy

$$\Pi(\mathbf{u}) = \frac{1}{2}\int_A \boldsymbol{\varepsilon}^T D\,\boldsymbol{\varepsilon}\; t\, dA \;-\; \int_{\Gamma_t} \mathbf{u}^T \bar{\mathbf{t}}\; t\, ds$$

- Among displacement fields that satisfy the clamp, the exact solution makes Π stationary ($\delta\Pi = 0$ is the weak form of equilibrium and the traction BCs); D is positive definite, so it is a minimum.
- FE approximation $\mathbf{u} \approx N\mathbf{d}$, $\boldsymbol{\varepsilon} = B\mathbf{d}$ in each element:

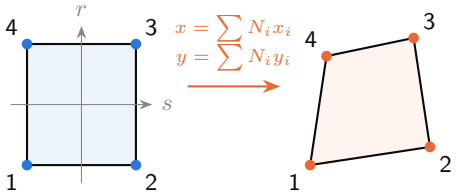
$$\Pi(\mathbf{d}) = \tfrac{1}{2}\,\mathbf{d}^T K \mathbf{d} - \mathbf{d}^T \mathbf{f}, \qquad K = \sum_e \int_{A_e} B^T D B\, t\, dA, \qquad \mathbf{f} = \sum_e \int_{\Gamma_e} N^T \bar{\mathbf{t}}\, t\, ds$$

$$\frac{\partial \Pi}{\partial \mathbf{d}} = 0 \quad \Longrightarrow \quad K\mathbf{d} = \mathbf{f} \qquad \text{(supports: section 5)}$$

A consequence used later (V6c)

At the solution $\mathbf{d}^T K \mathbf{d} = \mathbf{f}^T \mathbf{d}$, so $\Pi_{\min} = -\tfrac{1}{2}\,\mathbf{f}^T \mathbf{d}$: minimising Π is the same as maximising the compliance $\mathbf{f}^T \mathbf{d}$.

3. The Q4 element: shape functions and isoparametric mapping



Bilinear shape functions (nodes counter-clockwise)

$$N_i(s,r) = \tfrac{1}{4}(1 + s\,s_i)(1 + r\,r_i), \qquad s_i, r_i = \pm 1$$

Isoparametric: the same N_i interpolate geometry and displacement

$$x = \textstyle\sum_i N_i x_i, \quad y = \sum_i N_i y_i$$

$$u = \textstyle\sum_i N_i u_i, \quad v = \sum_i N_i v_i$$

Properties (tested)

- $N_i(s_j, r_j) = \delta_{ij}$: nodal values are the DOFs
- $\sum_i N_i = 1$: rigid translations are exact
- "bilinear": a product of two 1D linear functions

Code: fem/q4.py (shape_functions, shape_derivatives).

3. The Q4 element: Jacobian, chain rule, strain–displacement matrix

$$[J] = \begin{bmatrix} x_{,s} & y_{,s} \\ x_{,r} & y_{,r} \end{bmatrix} = \begin{bmatrix} N_{1,s} & \cdots & N_{4,s} \\ N_{1,r} & \cdots & N_{4,r} \end{bmatrix} \begin{bmatrix} x_1 & y_1 \\ \vdots & \vdots \\ x_4 & y_4 \end{bmatrix}, \qquad \begin{Bmatrix} N_{i,x} \\ N_{i,y} \end{Bmatrix} = [J]^{-1} \begin{Bmatrix} N_{i,s} \\ N_{i,r} \end{Bmatrix}$$

$$\boldsymbol{\varepsilon} = B\,\mathbf{d}_e, \qquad B = \begin{bmatrix} N_{1,x} & 0 & N_{2,x} & 0 & \cdots \\ 0 & N_{1,y} & 0 & N_{2,y} & \cdots \\ N_{1,y} & N_{1,x} & N_{2,y} & N_{2,x} & \cdots \end{bmatrix}$$

$$\mathbf{d}_e = [u_1, v_1, u_2, v_2, u_3, v_3, u_4, v_4]^T, \qquad [J]^{-1} = \frac{1}{\det J} \begin{bmatrix} y_{,r} & -y_{,s} \\ -x_{,r} & x_{,s} \end{bmatrix}$$

- $dA = \det J \; ds\, dr$; the code requires $\det J > 0$ (clockwise or badly distorted elements are rejected)
- Rectangle $2a \times 2b$: $J = \mathrm{diag}(a, b)$ and $\det J = ab =$ area$/4$ (tested)

Code: fem/q4.py (jacobian, strain_displacement); the same lines work for one element or for all elements at once.

3. The Q4 element: stiffness and Gauss quadrature

$$k = t \int_{-1}^{1}\int_{-1}^{1} B^T D B \, \det J \, ds \, dr \;\approx\; t \sum_{i=1}^{2}\sum_{j=1}^{2} B^T D B \, \det J \, w_i w_j$$

with the 2×2 Gauss points $s_i, r_j = \pm 1/\sqrt{3}$ and weights $w_i = w_j = 1$.

Why 2×2 Gauss is exact for a rectangle

An n-point rule is exact for polynomials of degree $\leq 2n-1$ (here 3). With $x = as$, $y = br$: $N_{i,x} = \frac{s_i(1+rr_i)}{4a}$ is linear in r only, $N_{i,y} = \frac{r_i(1+ss_i)}{4b}$ linear in s only, and $\det J = ab$. So every entry of $B^T D B$ is at most quadratic in s and in r: exact.

General quadrilateral: J^{-1} makes the integrand rational, so Gauss is approximate.

Closed form by hand (Cook §3.6):

$$k_{11} = \frac{Et}{1-\nu^2}\left[\frac{b}{3a} + \frac{(1-\nu)a}{6b}\right]$$

$$k_{12} = \frac{Et}{8(1-\nu)}$$

All 64 entries follow from $\int N_{i,x} N_{j,x} \, dA = \frac{b}{4a} s_i s_j \left(1 + \frac{r_i r_j}{3}\right)$ and two similar integrals. k depends only on a/b, not on the element size.

Code: fem/q4.py (gauss_rule, element_stiffness). Closed form: checks/closed_form.py. Proof that 2×2 Gauss is exact: checks/derivations.py.

3. The Q4 element: zero-energy modes, rank and completeness

- **Rigid-body modes.** Translations and the rotation $u = -\theta y,\ v = \theta x$ give $\varepsilon_x = \varepsilon_y = 0$, $\gamma_{xy} = -\theta + \theta = 0$, so $k\,\mathbf{d}_{\text{rigid}} = \mathbf{0}$: **3 zero eigenvalues**.
- **Rank.** Each Gauss point adds rank ≤ 3. With 2×2 the rank is $8 - 3 = 5$: no spurious modes. One point gives rank ≤ 3, i.e. 2 extra zero-energy "hourglass" modes. Hence full integration.
- **Completeness.** $\sum N_i = 1$, $\sum N_i x_i = x$, $\sum N_i y_i = y$ on *any* quadrilateral, so every linear field $u = a_0 + a_1 x + a_2 y$ is represented exactly: every constant strain is exact.

Why the patch test (V3) must pass on distorted meshes

$\det J\,[N_{i,x}, N_{i,y}]^T = \mathrm{adj}(J)\,[N_{i,s}, N_{i,r}]^T$ is *bilinear* in (s, r). For a constant stress the nodal forces $t \int B^T \boldsymbol{\sigma} \det J\, ds\, dr$ are therefore integrated exactly by 2×2 Gauss. They equal the consistent loads of the boundary tractions, so the exact linear field satisfies the discrete equations exactly.

4. Assembly and consistent loads

Assembly. Node n owns the global DOFs $2n$ (u) and $2n+1$ (v):

$$K = \sum_e L_e^T \, k_e \, L_e \qquad (L_e \text{ picks the element's 8 DOFs})$$

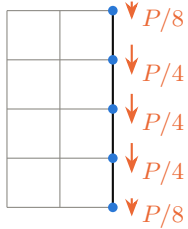
Code: 64 (row, column, value) triplets per element; COO $\rightarrow$ CSR adds the duplicates = this sum.

Consistent load of a uniform traction. On an edge of length h, $N_{1,2} = (1 \mp \xi)/2$ and $ds = \frac{h}{2} d\xi$:

$$f_i = t \int N_i \, \bar{\tau} \, ds = \frac{t \, h \, \bar{\tau}}{2} \quad \text{at each end of the edge}$$

With $\bar{\tau} = P/(tD)$ and $h = D/n_y$: P/n_y at the interior end nodes and $P/(2n_y)$ at the two corners; the sum is P (tested).

Code: fem/mesh.py, fem/assemble.py, fem/loads.py, fem/post.py (tip_deflection).



end face, $n_y = 4$

Tip deflection $v(L, 0)$: the mid-height end node (n_y even); for $n_y = 1$, linear interpolation along the edge (the mean of the two end nodes).

5. The clamp by Lagrange multipliers

Constraints, one row per clamped DOF ($m = 2(n_y + 1)$ rows; each row of C holds a single 1):

$$C\mathbf{d} = \mathbf{g} \qquad \text{(clamp: } \mathbf{g} = \mathbf{0}\text{; V1 and V3 also use } \mathbf{g} \neq \mathbf{0}\text{)}$$

Augmented functional

$$\Pi^*(\mathbf{d}, \boldsymbol{\lambda}) = \tfrac{1}{2}\,\mathbf{d}^T K \mathbf{d} - \mathbf{d}^T \mathbf{f} + \boldsymbol{\lambda}^T (C\mathbf{d} - \mathbf{g})$$

Stationarity

$$\frac{\partial \Pi^*}{\partial \mathbf{d}} = \mathbf{0} : \; K\mathbf{d} + C^T \boldsymbol{\lambda} = \mathbf{f}, \qquad \frac{\partial \Pi^*}{\partial \boldsymbol{\lambda}} = \mathbf{0} : \; C\mathbf{d} = \mathbf{g} \qquad \Longrightarrow \qquad \boxed{\begin{bmatrix} K & C^T \\ C & 0 \end{bmatrix} \begin{Bmatrix} \mathbf{d} \\ \boldsymbol{\lambda} \end{Bmatrix} = \begin{Bmatrix} \mathbf{f} \\ \mathbf{g} \end{Bmatrix}}$$

- Size $n + m$; all DOFs stay in the system (no renumbering, no deleted rows)
- K alone is singular (3 rigid-body modes); the augmented matrix is not

Code: fem/lagrange.py (constraint_matrix, augmented_matrix, solve_lagrange).

Indefinite, so no Cholesky

$\mathbf{x} = [\mathbf{0}; \boldsymbol{\mu}]$: $\mathbf{x}^T A \mathbf{x} = 0$ with $\mathbf{x} \neq \mathbf{0}$.
$\mathbf{x} = [C^T \boldsymbol{\mu}; -\alpha \boldsymbol{\mu}]$:
$\mathbf{x}^T A \mathbf{x} = \boldsymbol{\mu}^T C K C^T \boldsymbol{\mu} - 2\alpha \left\| C^T \boldsymbol{\mu} \right\|^2 < 0$ for large α.

So A is not positive definite: we use sparse LU with pivoting (SuperLU, minimum-degree ordering on $A + A^T$).

Nonsingular when C has full row rank and K is positive definite on the null space of C (the clamp removes all rigid-body modes). Then A has exactly n **positive and** m **negative** eigenvalues (checked on 3×1, 6×2, 12×4).

The multipliers are the reactions

First block row: $K\mathbf{d} + C^T \boldsymbol{\lambda} = \mathbf{f}$. Discrete equilibrium: $K\mathbf{d} = \mathbf{f} + \mathbf{R}$. Hence

$$\boxed{\mathbf{R} = K\mathbf{d} - \mathbf{f} = -C^T \boldsymbol{\lambda}}$$

Each row of C picks one DOF, so the reaction at the i-th clamped DOF is $-\lambda_i$: the force the constraint applies to enforce $C\mathbf{d} = \mathbf{g}$. No post-processing is needed.

5. Lagrange multipliers = elimination + reactions

Order the DOFs as free (f) and constrained (c), so $C = [\,0 \;\; I\,]$:

$$\begin{bmatrix} K_{ff} & K_{fc} & 0 \\ K_{cf} & K_{cc} & I \\ 0 & I & 0 \end{bmatrix} \begin{Bmatrix} \mathbf{d}_f \\ \mathbf{d}_c \\ \boldsymbol{\lambda} \end{Bmatrix} = \begin{Bmatrix} \mathbf{f}_f \\ \mathbf{f}_c \\ \mathbf{g} \end{Bmatrix}$$

- Third row: $\mathbf{d}_c = \mathbf{g}$
- First row: $\boxed{K_{ff}\,\mathbf{d}_f = \mathbf{f}_f - K_{fc}\,\mathbf{g}}$ — exactly the eliminated system
- Second row: $\boldsymbol{\lambda} = \mathbf{f}_c - K_{cf}\mathbf{d}_f - K_{cc}\mathbf{g} = -\mathbf{R}_c$ — the reactions

So

Lagrange multipliers give the elimination displacements *exactly*, plus the reactions, without removing any equation. Proved symbolically for a generic K (checks/derivations.py) and checked numerically against elimination (written only inside the tests).

5. Scaling the constraint rows and columns

$K \sim 10^{10}$ N/m but C holds ones. Multiply the constraint rows *and* columns by $s = \max \operatorname{diag} K$:

$$\begin{bmatrix} K & sC^T \\ sC & 0 \end{bmatrix} \begin{Bmatrix} \mathbf{d} \\ \boldsymbol{\lambda}_s \end{Bmatrix} = \begin{Bmatrix} \mathbf{f} \\ s\mathbf{g} \end{Bmatrix}, \qquad A_s = SAS, \quad S = \operatorname{diag}(I, sI)$$

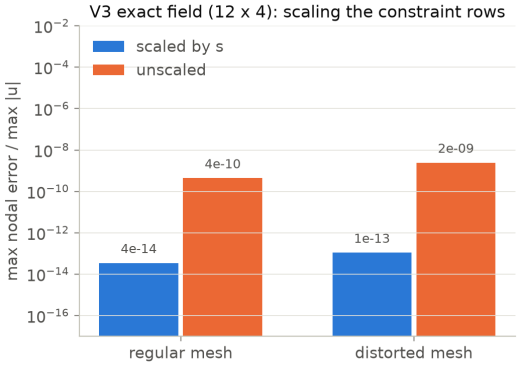
- Solution $S^{-1}[\mathbf{d}; \boldsymbol{\lambda}]$: $\mathbf{d}$ **unchanged**, $\boldsymbol{\lambda}_s = \boldsymbol{\lambda}/s$, and $\mathbf{R} = -(sC)^T\boldsymbol{\lambda}_s = -C^T\boldsymbol{\lambda}$
- A congruence, so the inertia is unchanged (Sylvester's law)
- Why $\sim 10^{22}$ unscaled: A^{-1} contains the block $K_{cf}K_{ff}^{-1}K_{fc} - K_{cc}$, as large as K, so $\|A\|\,\|A^{-1}\| \sim \|K\|^2$. Scaled, that block is divided by s^2, and $\operatorname{cond}(A_s) \approx \operatorname{cond}(K_{ff})$

cond_2	unscaled	scaled	K_{ff}
3×1	9.381×10^{21}	2.492×10^{4}	2.412×10^{4}
6×2	2.381×10^{22}	2.746×10^{5}	2.728×10^{5}
12×4	3.594×10^{22}	1.921×10^{6}	1.919×10^{6}
24×8	4.170×10^{22}	9.140×10^{6}	9.139×10^{6}

float64 SVD, confirmed by a 50-digit SVD on the two smallest meshes.

figures/exact_field.txt. Digit loss: Cook §9.3, Eq. 9.3-4: up to $\log_{10} \operatorname{cond}$ digits.

5. Scaling: the measured effect



figures/exact_field.txt, figures/exact_field_scaling.png.

- Exact linear field (V3, 12×4), relative nodal error: 4.4×10^{-10} **unscaled vs** 3.5×10^{-14} **scaled** (regular mesh), 2.5×10^{-9} vs 1.1×10^{-13} (distorted): about **four digits lost** without scaling
- Cook Eq. 9.3-4: up to $\log_{10} \mathrm{cond}$ digits may be lost: 22 unscaled (more than float64 has), 4–7 scaled
- Beyond $\mathrm{cond} \approx 1/\epsilon$ even eigenvalues are unreliable: on 12×4, `eigvalsh` of the unscaled matrix finds 134 positive / 6 negative instead of 130 / 10; the scaled matrix gives 130 / 10

6. Beam-theory reference

Moment (sagging positive): $M(x) = P(L - x)$.

Euler–Bernoulli: $EIv'' = M$, $v(0) = v'(0) = 0$:

$$v = \frac{P}{EI}\left(\frac{Lx^2}{2} - \frac{x^3}{6}\right) \;\Rightarrow\; v(L) = \frac{PL^3}{3EI}, \qquad I = \frac{tD^3}{12}$$

Stresses: $\sigma_{xx} = -\dfrac{My}{I} = -\dfrac{P(L-x)\,y}{I}$; equilibrium with $\tau_{xy}(\pm D/2) = 0$ gives

$$\tau_{xy} = \frac{P}{2I}\left(\frac{D^2}{4} - y^2\right), \qquad \int \tau_{xy}\, t\, dy = P$$

Timoshenko: shear strain $V/(\kappa GA)$, $\kappa = 5/6$ for a rectangle:

$$v(L) = \frac{PL^3}{3EI} + \frac{PL}{\kappa GA}$$

Values

$I = 2.8125 \times 10^{-5}$ m^4
Euler–Bernoulli: -4.74074 **µm**
shear term: -0.02080 µm (0.44%)
Timoshenko: -4.76154 **µm**
$\sigma_{xx}(L/2, \pm D/2) = \pm 26.6667$ kPa

Stresses are compared at mid-span, 6.7 depths from either end, where Saint-Venant end effects have decayed. Beam theory needs only the resultant P.

Values: `python -m checks.beam_theory`. All derived symbolically in checks/derivations.py. $\kappa = 5/6$: Cook p. 535.

7. Why Q4 is stiff in bending: parasitic shear (Cook §3.6)

Block in pure bending (Cook Eq. 3.6-9):

$$\varepsilon_x = -\frac{\theta y}{2a}, \quad \varepsilon_y = \nu\frac{\theta y}{2a}, \quad \gamma_{xy} = 0$$

Q4 bent the same way (sides stay straight, Eq. 3.6-10):

$$\varepsilon_x = -\frac{\theta y}{2a}, \quad \varepsilon_y = 0, \quad \gamma_{xy} = -\frac{\theta x}{2a}$$

Strain energies:

$$U_b = \frac{Et\,\theta^2 b^3}{6a}, \qquad U_{el} = \frac{t\,\theta^2 b}{6a}\left[\frac{Eb^2}{1-\nu^2} + Ga^2\right]$$

At equal moment $U = M\theta/2 = k\theta^2/2$, so $\theta_{el}/\theta_b = U_b/U_{el}$:

$$\boxed{\frac{\theta_{FE}}{\theta_{exact}} = \frac{1-\nu^2}{1 + \frac{1-\nu}{2}\left(\frac{a}{b}\right)^2}}$$

Cook Eq. 3.6-11, p. 99; a, b = element half-length and half-depth.

- $1-\nu^2$: from $\varepsilon_y = 0$
- $(a/b)^2$: only from the parasitic shear: **shear locking**
- $a/b = 1$: 0.674; our elements ($a/b = 4.44$): 0.115; one element for the whole beam ($a/b = 13.3$): 0.0144

Energies and Eq. 3.6-11 derived symbolically in checks/derivations.py; values from figures/bending_error.txt.

8. Convergence theory

Monotone compliance (minimum potential energy). Each mesh subdivides the previous one, so the displacement spaces are nested, $V_h \subset V_{h/2}$ (the coarse clamp is zero along the whole edge). A larger space cannot give a larger minimum: $\Pi_{h/2} \le \Pi_h$, and with $\Pi_{\min} = -\frac{1}{2}\mathbf{f}^T\mathbf{d}$:

$$\mathbf{f}^T\mathbf{d}_h \;\le\; \mathbf{f}^T\mathbf{d}_{h/2} \;\le\; \text{exact compliance}$$

The FE model is too stiff and can only soften under refinement.

Rate. Bilinear elements, smooth solution: energy error $O(h)$; displacements and outputs such as $v(L,0)$: $O(h^2)$.

Observed order from three nested meshes, $\phi = \phi_\infty + c\,h^q$:

$$\frac{\phi_1 - \phi_2}{\phi_2 - \phi_3} = 2^q \;\Rightarrow\; q = \log_2 \frac{\phi_1 - \phi_2}{\phi_2 - \phi_3}$$

Richardson extrapolation (Cook Eq. 9.7-1):

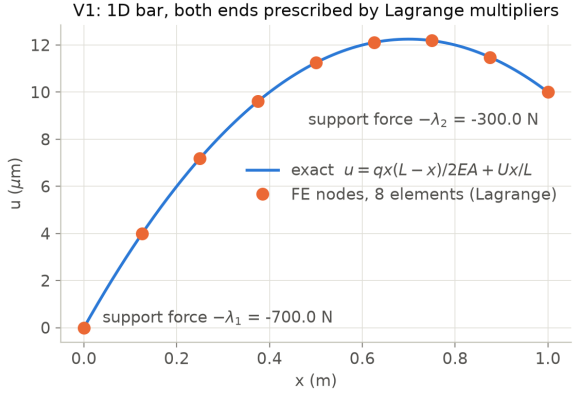
$$\phi_\infty = \frac{2^q\,\phi_{h/2} - \phi_h}{2^q - 1}$$

Valid for regular nested refinement with the quantity at a fixed node (both hold here).

Caveat: the clamped corners are singular points of the 2D solution and could lower the asymptotic rate on much finer meshes.

Code: fem/post.py (observed_order, richardson); formulas proved in checks/derivations.py.

V1: 1D bar, both ends prescribed — the Lagrange machinery alone



$EAu'' + q = 0$, $u(0) = 0$, $u(L) = U$:

$$u = \frac{qx(L-x)}{2EA} + \frac{Ux}{L}, \qquad N = EAu'$$

Support forces: $-N(0)$ and $+N(L)$.

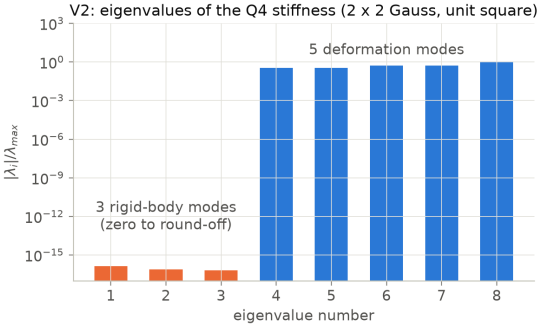
Linear elements with constant EA and consistent loads are **nodally exact** (Cook p. 47): the Green's function of a nodal load is piecewise linear, so Galerkin orthogonality makes the nodal error zero.

Max nodal error on 2, 3, 8, 32 elements: $\leq 4.8 \times 10^{-15}$ ✓

$-\lambda_1 = -700$ N, $-\lambda_2 = -300$ N = the exact support forces ✓

$L = 1$ m, $EA = 2 \times 10^7$ N, $q = 1000$ N/m, $U = 10^{-5}$ m. Same solver as the 2D problem. figures/bar1d.txt.

V2: the Q4 element stiffness

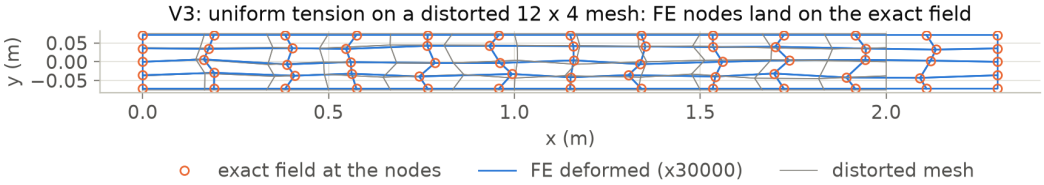


unit square	N/m
k_{11}, code	9.8901098901×10^9
k_{11}, formula	9.8901098901×10^9
k_{12}, code	3.5714285714×10^9
k_{12}, formula	3.5714285714×10^9

- All 64 entries equal exact integration to $\leq 2.3 \times 10^{-16}$ (4 rectangles) ✓
- 2×2 vs 3×3 Gauss: $\leq 7.7 \times 10^{-16}$ for rectangles, 8.3×10^{-3} for a general quadrilateral (integrand no longer polynomial)
- Exactly **3 zero eigenvalues** ($|\lambda|/\lambda_{\max} \leq 1.3 \times 10^{-16}$) and 5 deformation modes ✓

$t = 0.1$ m. figures/element_checks.txt.

V3: uniform tension — an exact linear field (patch test)



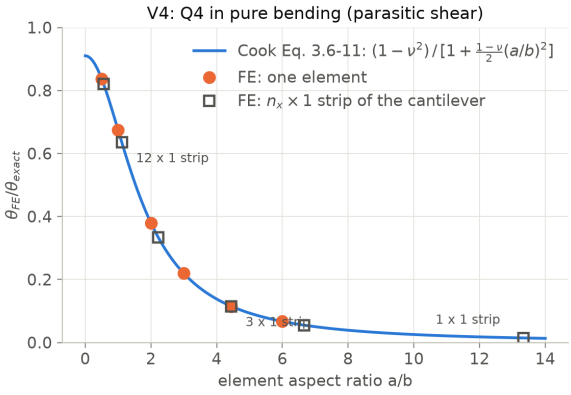
$u = \sigma_0 x/E$, $v = -\nu\sigma_0 y/E$ gives $\boldsymbol{\sigma} = [\sigma_0, 0, 0]$ everywhere: equilibrium, free top and bottom, $u = 0$ on $x = 0$, $v = 0$ at $(0, 0)$. All supports by Lagrange multipliers. Q4 contains every linear field, so the FE answer must be **exact** on any mesh.

12×4	nodal error	stress error
regular	3.5×10^{-14}	2.6×10^{-14}
distorted	1.1×10^{-13}	4.9×10^{-14}

Reactions = minus the consistent loads of σ_0 on the left edge ✓
Also with $\mathbf{g} \neq \mathbf{0}$ (exact field plus a rigid motion) ✓

figures/exact_field.txt; tests/test_exact_field.py.

V4: Q4 in pure bending against Cook Eq. 3.6-11



figures/bending_error.txt.

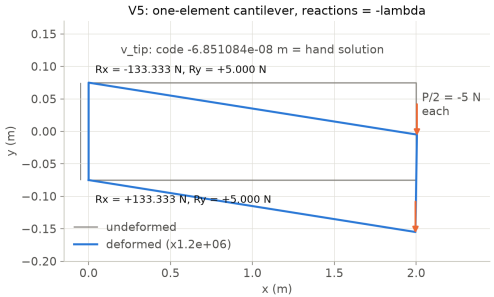
One clamped element under an end couple, and $n_x \times 1$ strips of the cantilever. Every element of a strip carries the same moment, so a strip behaves *exactly* like one element with $a = L/(2n_x)$.

case	a/b	$\theta_{FE}/\theta_{exact}$
one element	1	0.6740740741
3×1 strip	4.44	0.1149921997
1×1 strip	13.3	0.0143936731

FE = formula to $\leq 8.6 \times 10^{-13}$ in all 12 cases ✓

This is why the coarse meshes are so stiff.

V5: one-element cantilever, solved completely by hand



Symmetry (u odd, v even in y) leaves 2 unknowns. Minimising $\Pi(q, w)$:

$$v_{tip} = \frac{4PL(1+\nu)\left[2L^2(1-\nu)+D^2\right]}{EtD\left[L^2(1-\nu)+2D^2\right]}$$

$\sum F_x = 0$, $\sum M = 0$: $R_x = -PL/D$ (bottom), $+PL/D$ (top); $\sum F_y = 0$ *plus symmetry*: $R_y = -P/2$ each.

v_{tip} (code = hand)	$-6.8510837727 \times 10^{-8}$ m
u top / bottom	$\pm 3.8383128295 \times 10^{-9}$ m
R_x bottom / top	$+133.333$ / -133.333 N
R_y each	$+5.000$ N

Code = hand solution to $\leq 1.5 \times 10^{-13}$, and $\boldsymbol{\lambda} = -\mathbf{R}$ ✓

v_{tip} is only 0.01445 of Euler–Bernoulli: locking.

figures/one_element.txt. checks/derivations.py re-derives it over all four free DOFs, without assuming symmetry.

V6 (a, b): statics and Clapeyron on every mesh

mesh	$\sum R_x$ (N)	$\sum R_y$ (N)	moment of $\mathbf{R}$ (N m)	Clapeyron, relative
3×1	1.1×10^{-12}	9.999999999979	19.999999999970	1.6×10^{-13}
12×4	-2.3×10^{-11}	9.999999999496	19.999999999379	3.1×10^{-11}
48×16	9.2×10^{-12}	9.999999998828	19.999999999391	2.2×10^{-11}
192×64	2.2×10^{-10}	10.000000007987	20.000000008089	-7.3×10^{-10}
384×128	1.1×10^{-9}	9.999999980917	19.999999961142	1.4×10^{-9}
theory	0	$-P = 10$	$-PL = 20$	0

Moment about $(0, 0)$. Clapeyron: $(\mathbf{d}^T K \mathbf{d} - \mathbf{f}^T \mathbf{d})/\mathbf{f}^T \mathbf{d}$.

Why statics is exact

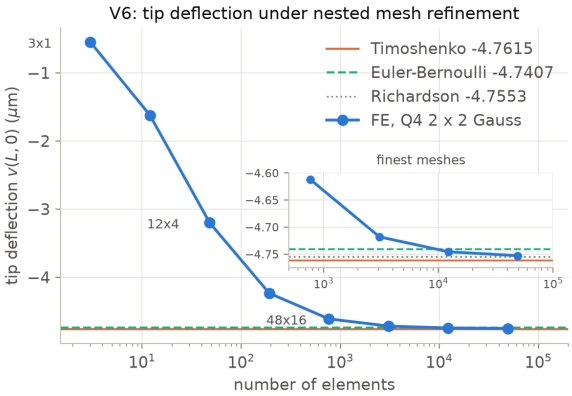
Each k_e annihilates rigid-body motions, so the nodal forces $K\mathbf{d}$ are self-equilibrated, and the reactions $\mathbf{R} = -C^T \boldsymbol{\lambda}$ balance the load exactly (to round-off).

Why Clapeyron holds

$\mathbf{d}^T(K\mathbf{d} + C^T\boldsymbol{\lambda}) = \mathbf{d}^T\mathbf{f}$ and $C\mathbf{d} = \mathbf{0}$, so $\mathbf{d}^T K \mathbf{d} = \mathbf{f}^T \mathbf{d}$: the reactions do no work.

All 8 meshes in figures/reactions.txt; the round-off grows with the condition number, as expected.

V6 (c, d): tip deflection under nested refinement



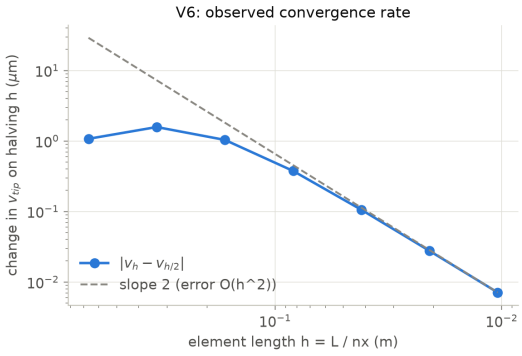
figures/tip_deflection.txt.

mesh	v_{tip} (μm)	/ E–B	/ Tim.
3×1	-0.54734	0.1155	0.1149
6×2	-1.62173	0.3421	0.3406
12×4	-3.19979	0.6750	0.6720
24×8	-4.23595	0.8935	0.8896
48×16	-4.61223	0.9729	0.9686
96×32	-4.71813	0.9952	0.9909
192×64	-4.74577	1.0011	0.9967
384×128	-4.75287	1.0026	0.9982

Compliance $\mathbf{f}^T\mathbf{d}$ increases on every refinement ✓

Equal to an independent implementation to all 5 printed digits ✓

V6 (e): observed convergence rate and Richardson extrapolation



finest mesh	order q	Richardson (µm)
48×16	1.461	-4.82679
96×32	1.829	-4.75961
192×64	1.938	-4.75553
384×128	1.961	-4.75532

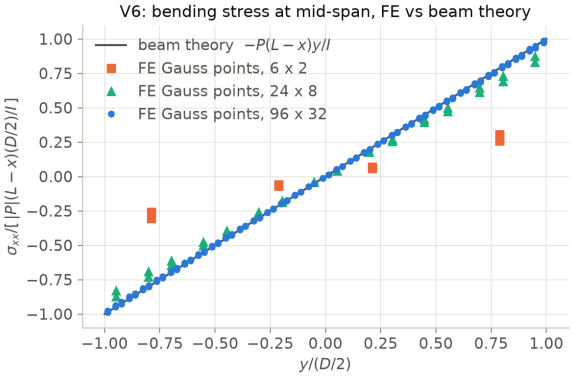
The order rises towards the theoretical 2 ✓
(coarse meshes: locking dominates).

Converged tip deflection

-4.75532 µm = **0.99869** × **Timoshenko** (-0.13%)
= 1.00308 × Euler–Bernoulli ($+0.31\%$)
The last two estimates differ by 0.004%.

Each row uses three nested meshes ending at the one named (e.g. 48×16: 12×4, 24×8, 48×16). figures/tip_deflection.txt.

V6 (d): bending stress at mid-span



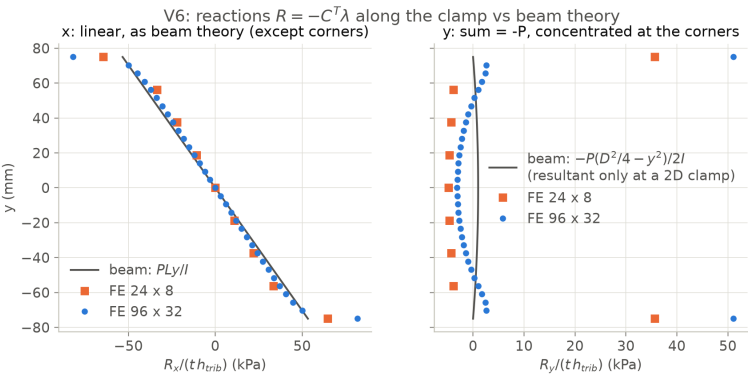
Gauss points of the element column at $x = L/2$, against $\sigma_{xx} = -P(L-x)y/I$ at the same points. Error: the largest $|\sigma_{FE} - \sigma_{beam}|$ over those points, divided by the beam surface stress at the same x.

mesh	error
3×1 / 6×2	0.5162 / 0.5308
24×8	0.1163
96×32	0.0150
384×128	0.0025

Falls on every refinement from 6×2 on ✓ (the two coarsest meshes are dominated by locking).

figures/midspan_stress.txt (all 8 meshes; equal to the independent table where it exists).

V6: reactions $\mathrm{R} = -C^T\boldsymbol{\lambda}$ along the clamp



figures/reactions.txt, figures/reactions.png.

R_x: linear, as beam theory (PLy/I); interior nodes within 5.4% on 96×32.

Corner nodes: $t_x/(PLy/I) = 1.211$, 1.536, 2.130 on 24×8, 96×32, 384×128; it keeps growing.

R_y: the sum is exactly $-P$, but it is concentrated at the corners. At a 2D clamp the parabola fixes only the resultant (Saint-Venant).

The clamped corners are **singular points** of 2D elasticity.

Conclusions

- **Lagrange multipliers behave exactly as the theory says**: the same displacements as elimination, and the multipliers *are* the reactions, $\mathbf{R} = -C^T\boldsymbol{\lambda}$; statics holds to round-off on every mesh. Non-zero prescribed values ($\mathbf{g} \neq \mathbf{0}$) need no change to the code.
- **Scaling is essential**: unscaled, $\mathrm{cond}_2 \approx 10^{22}$ and about four digits are lost on an exact field; scaling by $\max \mathrm{diag}\, K$ makes the system as well conditioned as the eliminated one.
- **Q4 locks in bending**: one element gives 0.01445 of the Euler–Bernoulli deflection and 3×1 gives 0.1155; in pure bending the error is exactly Cook Eq. 3.6-11.
- **Converged answer**: -4.75532 µm $= 0.99869\times$ Timoshenko $= 1.00308\times$ Euler–Bernoulli. Beam theory is not the exact 2D solution: the full clamp restrains warping and Poisson contraction at the root, and the load is a uniform traction. These effects scale like $(D/L)^2$, the same order as the shear term (0.44%), so a 0.1–0.3% gap is expected.
- **Every number has an analytical counterpart**, computed from its formula and checked by 109 tests; every closed form is derived symbolically.

Limitations



Model

- Linear elasticity, small displacements, plane stress
- One element type (Q4, full integration): fine meshes needed in bending (within 1% of Timoshenko only from 96×32 on)
- Structured rectangular meshes, one load case

Interpretation

- Stresses at the clamped corners are mesh-dependent (singularity, not resolved)
- The exact 2D solution for this clamp and load is outside the scope: beam theory is the reference, and the remaining gap is explained, not removed
- The order is observed for h from $L/12$ to $L/384$; the corner singularity could lower it on much finer meshes

Reproduce everything

`python -m pytest -q`	the 109 tests
`python run_all.py`	every figure and table in figures/
`python -m checks.derivations`	the symbolic proofs

Thank you

Questions?

Backup slides: code map ▪ symbolic proofs ▪ the V5 hand solution ▪ why the unscaled condition number is $\sim 10^{22}$

Backup: code map (equation → function → test)

Equation / step	Code	Test
Plane-stress D	`fem/material.py`	`test_q4.py`
N, $[J]$, B, k by Gauss	`fem/q4.py`	`test_q4.py` (V2, V4)
Mesh, DOF numbering	`fem/mesh.py`, `assemble.py`	all 2D tests
$K = \sum L^T k\, L$ (COO → CSR)	`fem/assemble.py`	`test_cantilever.py`
Consistent edge loads	`fem/loads.py`	`test_cantilever.py`, `test_exact_field.py`
C, scaled solve, $\mathbf{R} = -C^T \boldsymbol{\lambda}$	`fem/lagrange.py`	`test_bar1d.py`, `test_cantilever.py`
Cantilever driver	`fem/cantilever.py`	V5, V6 tests
Stresses, $v(L, 0)$, resultants, order, Richardson	`fem/post.py`	V3, V6 tests
Closed forms V2–V5	`checks/closed_form.py`	`test_q4.py`, `test_one_element.py`
Beam theory	`checks/beam_theory.py`	V4, V6 tests
Symbolic proofs	`checks/derivations.py`	`test_derivations.py`

Backup: what `checks/derivations.py` proves (sympy)

- Plane-stress D from 3D Hooke's law
- Exact k of a rectangle; 2×2 Gauss reproduces it exactly; k_{11}, k_{12}
- Cook Eq. 3.6-11 from the two strain energies
- One-element cantilever: all displacements and all four reactions, minimising Π over the four free DOFs (no symmetry assumed)
- Euler–Bernoulli by integrating $EIv'' = M$; Timoshenko shear term; $I = tD^3/12$
- τ_{xy} from equilibrium, and $\int \tau_{xy} t\, dy = P$
- Bar: ODE, end conditions, support forces
- Uniform tension: $\boldsymbol{\sigma} = [\sigma_0, 0, 0]$ and the boundary conditions
- Consistent edge load $th\bar{\tau}/2$
- Lagrange = elimination; $\boldsymbol{\lambda} = -\mathbf{R}$; $\boldsymbol{\lambda}_s = \boldsymbol{\lambda}/s$ under scaling
- Richardson's formula; $(\phi_1 - \phi_2)/(\phi_2 - \phi_3) = 2^q$

How it is used

Each derived expression is compared with the expression the code uses; `tests/test_derivations.py` fails if any pair differs.

Backup: the V5 hand solution in detail

Nodes 1 $(0,-b)$, 2 $(L,-b)$, 3 (L,b), 4 $(0,b)$; $a=L/2$, $b=D/2$; nodes 1 and 4 clamped; $P/2$ on v at 2 and 3. Symmetry: $u_3=-u_2=q$, $v_2=v_3=w$.

$$u = q\,r\,\frac{1+s}{2}, \quad v = w\,\frac{1+s}{2} \;\Rightarrow\; \varepsilon_x = \frac{qr}{2a}, \;\; \varepsilon_y = 0, \;\; \gamma_{xy} = \frac{q(1+s)}{2b} + \frac{w}{2a}$$

$$\Pi(q,w) = \frac{t\,ab}{2}\int_{-1}^{1}\!\!\int_{-1}^{1} \left(D_{11}\varepsilon_x^2 + G\gamma_{xy}^2\right) ds\,dr - Pw$$

$$\frac{\partial\Pi}{\partial q} = 0, \;\; \frac{\partial\Pi}{\partial w} = 0 \;\Rightarrow\; q = -\frac{3a^2P}{t(Ga^2 + D_{11}b^2)}, \qquad w = \frac{aP(D_{11}b^2 + 4Ga^2)}{tGb(Ga^2 + D_{11}b^2)}$$

With $D_{11} = E/(1-\nu^2)$, $G = E/(2(1+\nu))$, $a=L/2$, $b=D/2$ this is the v_{tip} formula of V5, and $u_{top} = -u_{bottom} = -\dfrac{6PL^2(1-\nu^2)}{Et\,[L^2(1-\nu)+2D^2]}$.

Reactions: $\sum F_x = 0$ and $\sum M_O = 0$ give $R_{x1} + R_{x4} = 0$ and $PL + b(R_{x1} - R_{x4}) = 0$, so $R_{x1} = -PL/D$; $\sum F_y = 0$ and symmetry give $R_{y1} = R_{y4} = -P/2$.

Backup: why the unscaled condition number is $\sim 10^{22}$

With $C = [\,0 \;\; I\,]$ (scaled by s), the exact inverse of the saddle-point matrix is, in the order $[\mathbf{d}_f, \mathbf{d}_c, \boldsymbol{\lambda}_s]$:

$$A_s^{-1} = \begin{bmatrix} K_{ff}^{-1} & 0 & -K_{ff}^{-1}K_{fc}/s \\ 0 & 0 & I/s \\ -K_{cf}K_{ff}^{-1}/s & I/s & \left(K_{cf}K_{ff}^{-1}K_{fc} - K_{cc}\right)/s^2 \end{bmatrix}$$

- The last block is minus the stiffness the supports see: as large as K ($\sim 10^{11}$ N/m)
- $s = 1$: $\|A\| \sim \|K\| \sim 10^{11}$ and $\|A^{-1}\| \gtrsim 10^{11}$, so $\mathrm{cond} \sim 10^{22}$
- $s = \max \mathrm{diag}\, K$: that block shrinks to $\sim 1/s$ and the coupling blocks are divided by s; $\|A_s^{-1}\| \approx \|K_{ff}^{-1}\|$, so $\mathrm{cond}(A_s) \approx \mathrm{cond}(K_{ff})$ (measured within 3%)

Checked numerically with this exact inverse and with a 50-digit SVD (mpmath) on 3×1 and 6×2.